

ILOPANGO, El Salvador

WMO: 786630

Lat: 13.699N

Lon: 89.120W

Elev: 2021

StdP: 13.65

Time Zone: -6.00 (W06)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	62.2	63.4	51.8	61.6	71.4	54.3	67.5	71.4	20.5	78.3	18.8	78.5	8.1	0	0.261

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.6	91.6	73.6	90.0	73.9	89.4	74.1	79.1	86.4	78.3	85.6	77.7	85.0	8.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	152.4	84.6	76.2	147.3	83.6	75.4	143.7	82.9	44.6	86.2	43.5	86.5	42.8	85.4	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	16.0	13.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.3	76.1	77.3	78.7	79.3	78.5	77.5	77.8	77.6	77.0	76.4	76.1	75.7
	DBStd	2.11	2.03	1.79	1.85	1.91	1.67	1.72	1.68	1.41	1.61	1.93	1.94	2.09
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9975	809	766	891	878	883	824	862	855	809	818	783	797
	CDD65	4500	344	346	426	428	418	374	397	390	359	353	333	332
	CDH74	31020	2593	2713	3514	3305	2802	2324	2817	2589	2002	1897	2148	2316
	CDH80	11010	1027	1143	1546	1292	900	696	1002	863	557	520	676	788
(14) Wind	WSAvg	6.0	7.7	6.9	7.1	6.5	5.1	4.6	5.0	5.0	4.7	5.1	7.2	7.4
(15) Precipitation	PrecAvg	68.2	0.2	0.2	0.6	1.8	6.0	11.6	12.9	12.7	13.1	8.6	1.3	0.4
	PrecMax	78.4	1.9	2.3	2.8	5.0	11.0	17.4	16.5	17.3	20.5	15.0	5.0	3.9
	PrecMin	58.4	0.0	0.0	0.0	0.0	1.3	6.2	9.1	9.1	5.4	3.3	0.2	0.0
	PrecStd	5.9	0.4	0.5	0.7	1.6	2.2	3.4	2.9	2.4	4.1	3.8	1.3	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.3	92.7	93.5	93.1	90.6	89.4	90.0	89.5	88.2	88.1	88.7	89.5
		MCWB	75.0	73.5	72.7	74.0	75.5	77.5	76.8	76.7	77.7	77.5	77.1	74.9
	2%	DB	89.5	90.5	91.6	90.8	88.2	87.6	88.5	87.9	86.8	86.3	87.3	87.8
		MCWB	73.4	73.5	72.0	74.2	76.2	76.7	76.0	76.4	77.2	76.5	75.5	73.4
	5%	DB	87.9	89.2	89.9	89.3	86.9	86.0	87.5	86.4	85.6	84.6	85.9	86.2
		MCWB	72.7	72.8	71.9	74.0	76.3	76.5	75.7	76.4	76.7	75.6	74.6	72.5
	10%	DB	86.1	87.4	88.1	87.5	85.5	84.3	85.9	85.5	83.9	83.4	84.2	84.6
		MCWB	72.0	72.0	72.0	73.7	76.2	76.1	76.0	76.3	76.1	75.1	73.7	71.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.5	78.2	77.6	78.9	79.1	79.2	79.4	79.6	79.8	79.6	78.9	77.8
		MCDB	88.1	87.8	87.6	87.2	86.8	86.2	86.5	86.4	86.8	86.2	87.0	86.4
	2%	WB	75.2	76.4	76.0	77.5	78.0	78.0	78.2	78.3	78.2	77.9	77.1	75.7
		MCDB	87.0	86.6	87.1	86.6	85.6	85.0	85.7	85.3	84.9	84.2	84.5	84.9
	5%	WB	74.2	75.0	74.9	76.4	77.2	77.2	77.4	77.4	77.1	76.8	76.0	74.2
		MCDB	85.7	85.8	86.1	85.4	84.6	84.1	84.7	84.3	83.3	82.8	83.1	83.7
	10%	WB	73.0	73.7	73.8	75.2	76.4	76.4	76.5	76.6	76.2	75.7	74.7	72.9
		MCDB	84.1	84.4	84.2	83.6	83.3	82.7	83.5	83.3	81.9	81.4	81.8	82.1
(35) Mean Daily Temperature Range	5% DB	MDBR	22.5	22.8	22.5	18.6	15.8	15.4	17.3	17.0	15.3	15.1	18.2	20.3
		MCDBR	23.9	24.4	24.0	20.6	17.8	17.5	19.1	18.6	17.6	17.2	19.5	21.2
		MCWBR	11.4	10.9	10.4	8.9	8.6	9.0	9.5	10.0	9.4	9.3	10.4	10.4
	5% WB	MCDBR	23.0	23.4	22.1	18.8	16.3	16.4	17.9	17.6	16.4	15.9	18.3	20.6
		MCWBR	11.6	11.9	10.6	9.0	8.3	8.9	9.9	9.9	9.2	9.1	10.4	11.1
(40) Clear-Sky Solar Irradiance	taub		0.344	0.356	0.390	0.477	0.502	0.415	0.417	0.411	0.405	0.383	0.362	0.345
	taud		2.588	2.554	2.435	2.178	2.133	2.428	2.424	2.432	2.435	2.538	2.571	2.605
	Ebn at Noon		304	303	293	267	256	276	276	281	285	291	295	301
	Edh at Noon		30	33	38	49	50	37	37	37	37	33	31	30
(44) All-Sky Solar Radiation	RadAvg		1770	1948	2056	1966	1768	1784	1988	1955	1746	1641	1722	1702
	RadStd		41	64	106	111	129	100	114	129	127	140	64	44

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page